BINALI ASSALAARACHCHI

AI & SOFTWARE ENGINEERING UNDERGRADUATE

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PROFESSIONAL SUMMARY

Third-year BSc (Hons) Information Technology undergraduate at SLIIT, specializing in Artificial Intelligence, seeking a Full Stack AI Engineer internship. Hands-on experience building and deploying full-stack applications, integrating AI APIs, developing machine learning workflows, and working with web and mobile technologies.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C#

Frontend & Backend: React.js, Node.js, Express.js, Django, ASP.NET, REST APIs, HTML, CSS

AI / ML: Scikit-learn, Pandas, NumPy, data preprocessing, feature engineering, model training, model evaluation, cosine similarity, heuristic search, AI API integration

Databases & Cloud: PostgreSQL, MySQL, MongoDB, Firebase/Firestore

Mobile & Tools: Flutter, React Native, Git, GitHub, API integration, deployment, OOP, MVC, Agile/Scrum

SOFTWARE ENGINEERING PROJECTS

CeylonTrailAI — AI-Powered Travel Platform | *Team Project · Year 3, Semester 1*

Technologies: ASP.NET, C#, PostgreSQL, Flutter

- Implemented Travel Alerts CRUD functionality for creating, retrieving, updating, and deleting travel alerts.
- Worked on the Travel Intelligence component and Validation Agent within the team's travel-focused application.
- Contributed to a solution combining backend services, database functionality, and a Flutter mobile application.

Driver Dashboard — Web Platform | *Team Project · Year 2, Semester 2*

Technologies: React.js, Node.js, MySQL, Python AI component

- Contributed to a full-stack travel itinerary planning and vehicle booking system with an AI-powered itinerary recommendation engine.
- Worked on frontend development, API integration, database schema design, and deployment.

Driver Dashboard — Mobile App | *Team Project · Year 2, Semester 2*

Technologies: React Native, Node.js, MongoDB

- Developed the mobile companion app for the travel-planning platform and integrated it with backend services.

Laundry Management System — Payment Module & Database | *Team Project · Year 2, Semester 1*

Technologies: Java, Spring Boot, Maven, MySQL, HTML/CSS/JavaScript

- Designed and implemented the payment and database module, applying MVC and relational database design.

Order Management Component — Inventory Management System | *Team Project · Year 1, Semester 2*

Technologies: Java, OOP

- Built the order management component using encapsulation, inheritance, and polymorphism.

AI / ML PROJECTS

Lunelle — PMDD AI Companion | *Individual Project · Completed & Deployed*

Technologies: React, Node.js, Express.js, Firebase/Firestore, Gemini API

- Built and deployed a mobile-first application for symptom tracking, cycle-based pattern awareness, and personalized non-diagnostic insights.
- Integrated server-side Gemini API functionality and structured symptom processing, with clear boundaries that the app is not a diagnostic tool.

Live demo: https://lunelle-pmdd-ai.web.app/welcome

Pac-Man Search Algorithms | *Intelligent Agents Coursework*

Technologies: Python

- Implemented and explored BFS, DFS, and A* search, including heuristic functions for pathfinding tasks.

Machine Learning Model Development | *Coursework*

Technologies: Python, Pandas, NumPy, Scikit-learn

- Carried out an end-to-end machine learning workflow including data preprocessing, feature engineering, model training, and evaluation.

CineMatch — AI-Powered Movie Recommender | *Individual Project*

Technologies: Python, Django, Scikit-learn, Pandas, React.js, TMDB API, MySQL

- Developed a content-based movie recommender using cosine similarity, with a Django REST API and React interface.
- Integrated TMDB data and JustWatch streaming availability.

Student Performance Prediction & Study Schedule Generator | *Individual Project*

Technologies: Python, Django, Scikit-learn, Flutter, MySQL

- Developed an ML pipeline to predict student performance and connected model outputs to a Flutter interface through a REST API.
- Generated personalized study plans from prediction outputs.

Heart Disease Prediction — ML Model | *Team Project*

Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

- Performed data cleaning and feature selection; compared Logistic Regression, Random Forest, and SVM classifiers.
- Evaluated model performance using accuracy, precision, and recall.

EDUCATION

BSc (Hons) Information Technology — Specialization in Artificial Intelligence | SLIIT

Year 3 | CGPA: 3.58

BCS Professional Graduate Diploma in IT | British Computer Society (BCS) | Graduated 2026

Certificate in Computer Science | National Institute of Business Management (NIBM) | 2021–2022

GCE Advanced Level — Commerce Stream | St. Paul's Girls' School | 2021–2023 | Accounting (A), Economics (A), ICT (A), General English (A)

LEADERSHIP & INVOLVEMENT

Member (Volunteer) — Leo Club of SLIIT | 2025–Present

Organizing Committee Vice President (Marketing) — Kokila Rawaya '25, AIESEC in SLIIT

MXP Marketing Team Leader — AIESEC in SLIIT | Jul 2025–Jan 2026

Campaign Coordinator, Outgoing Global Teacher (OGTe) — AIESEC in SLIIT | Jul 2025–Jan 2026

CERTIFICATIONS & ACHIEVEMENTS

- SLIIT Faculty of Computing Dean's List — Year 2, Semester 2
- Deloitte Australia Data Analytics Job Simulation — Forage
- AI Skills Fest 2026 — Microsoft | Jun 2026
- SLIIT Faculty of Computing Dean's List — Year 1, Semester 2
- Emerging Marketer — Legacy 2025, AIESEC in SLIIT (Nominee)
- British Council Key English Test (KET) — Pass with Merit

REFERENCES

Umayangana Wijayasiri

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